



THE
ZSMC CO
ESTD 2026



Research Data & Reproducibility Policy

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Effective Date: 16-08-2026

Evidence before prestige. Reproducibility before publicity. Intellectual contribution before title. Scientific honesty before institutional interest.

This policy establishes standards for the collection, management, preservation, analysis, documentation, and dissemination of research data and computational materials produced by The ZSMC Co.

Its purpose is to ensure that ZSMC research is sufficiently documented to permit meaningful scrutiny, reproduction, reuse, or independent verification where reasonably practicable.

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Founder
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Scope

This policy applies to research data and associated materials generated or substantially used in ZSMC research, including:

- experimental measurements;
- observational data;
- simulation outputs;
- computational results;
- source code;
- benchmark datasets;
- laboratory records;
- protocols;
- analysis scripts;
- models;
- figures and derived datasets; and
- research documentation.

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Data Integrity

Research data shall be recorded as accurately as reasonably practicable.

Original observations and measurements should be preserved in a form that allows later verification of their provenance.

Researchers shall not deliberately alter, fabricate, selectively discard, or manipulate data in a manner intended to misrepresent research findings.

Data transformations and processing steps should be documented sufficiently to permit reconstruction of significant results.

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Raw and Derived Data

Where applicable, ZSMC research shall distinguish between:

- **raw data:** measurements or observations substantially as originally obtained;
- **processed data:** data subjected to documented transformations;
- **derived data:** values calculated or inferred from other data; and
- **reported results:** values or conclusions presented in a research output.

The relationship between these stages should be documented for significant research outputs.

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Experimental Records

Experimental work should record, as appropriate:

- date and time;
- researcher;
- experiment or sample identifier;
- materials and relevant batch information;
- equipment and relevant settings;
- environmental conditions where material;
- experimental procedure;
- deviations from the planned procedure;
- observations;
- measurements;
- failures or anomalies; and
- conclusions or follow-up actions.

Experimental records should be sufficiently detailed to permit another competent researcher to understand what was performed.

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Computational Research

Computational research should, where reasonably practicable, preserve:

- source code;
- software versions;
- dependencies;
- configuration information;
- input data;
- relevant random seeds;
- computational parameters;
- hardware or execution environment where material;
- analysis scripts; and
- procedures necessary to reproduce reported results.

Where exact reproduction is impossible because of proprietary, unavailable, or changing dependencies, the limitation should be disclosed.

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Version Control

Important research code, documents, and computational workflows should use version control where practical.

Material changes should be traceable through version history.

Published or formally reported computational results should identify the relevant version of software or research code where appropriate.

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Data Storage and Preservation

Research materials shall be stored in a manner reasonably designed to prevent accidental loss, unauthorized alteration, or corruption.

Important research outputs should have appropriate backups.

Where feasible, significant research datasets and source materials should be preserved beyond the immediate completion of the project.

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Public Release

ZSMC shall seek to make research materials publicly available where doing so is scientifically useful and legally and ethically permissible.

Depending on the project, this may include:

- source code;
- datasets;
- protocols;
- supplementary material;
- technical reports;
- experimental records in suitable form; and
- reproducibility documentation.

Public release is not required where disclosure would create legitimate safety, privacy, legal, security, intellectual-property, or contractual concerns.

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Intellectual Property and Confidentiality

This policy does not require disclosure of information that ZSMC is legally or contractually required to keep confidential.

Where intellectual-property protection is reasonably contemplated, public disclosure shall be coordinated appropriately before dissemination.

Restrictions on public release should not be used merely to conceal inconvenient scientific results.

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Reproducibility Standards

Researchers shall make reasonable efforts to ensure that significant reported results can be independently evaluated.

The level of documentation required should be proportionate to the significance and complexity of the research.

For important claims, researchers should seek, where feasible:

- repeated measurements;
- independent runs;
- replicate experiments;
- sensitivity analysis;
- uncertainty estimation;
- independent computational verification; or
- other appropriate validation.

The appropriate standard will vary by research field.

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Failed and Negative Results

Material failed experiments, anomalous results, and negative findings should be retained where they provide meaningful information about the research process or conclusions.

Failure to obtain a desired result shall not by itself constitute a reason to discard the underlying record.

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Data Availability Statements

Research outputs should include an appropriate statement describing the availability of relevant data, code, protocols, or other research materials where required or useful.

Where materials cannot be publicly released, the reason should be stated where appropriate.

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Reproducibility and Publicity

Publicity shall not precede the minimum level of verification reasonably appropriate to the claim.

Preliminary results may be communicated as preliminary results, but shall not be represented as established findings.

The desire for publicity, recognition, funding, or institutional reputation shall not determine whether evidence is disclosed or characterized accurately.

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Corrections

If an error is discovered in data, code, documentation, or a published result, the relevant record shall be corrected where appropriate.

Where an error materially affects a published conclusion, ZSMC shall seek to notify relevant collaborators, repositories, publishers, or readers as appropriate.

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Responsibilities

Researchers are responsible for:

- maintaining appropriate research records;
- preserving relevant original data;
- documenting significant transformations;
- maintaining reproducible computational workflows where applicable;
- identifying limitations;
- safeguarding research materials;
- complying with applicable legal, ethical, safety, and contractual requirements; and
- making appropriate materials available when permissible.

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Review

This policy shall be reviewed as ZSMC's research methods, infrastructure, and data requirements develop.

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